



OPEN Evaluation and telemetry-based detection of GPS spoofing effects on UAV navigation using software-defined radio

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This paper presents an experimental investigation into the vulnerabilities of autonomous drone navigation systems under GPS spoofing attacks using Software Defined Radio (SDR). A low-cost spoofing framework was implemented using HackRF One and GPS-SDR-SIM to manipulate the navigation signals of an F450 quadcopter equipped with a Pixhawk 4 flight controller and a Ublox NEO-M8N GPS module. Both static and dynamic spoofing trials were conducted in controlled outdoor environments. During normal autonomous waypoint missions, the drone demonstrated stable sensor outputs and accurate trajectory tracking. However, under spoofed GPS conditions, the drone deviated from its intended path, breached geofenced boundaries, and exhibited unstable behaviour leading to an uncontrolled descent and crash landing instead of executing the intended Return-to-Launch (RTL) response. Detailed analysis of barometric altitude, yaw angle, magnetometer, vibration data, as well as GPS indicators such as satellite count and HDOP, revealed significant anomalies during spoofed flights. To enhance interpretability, a telemetry-based detection indicator was introduced by computing deviation thresholds between GPS, IMU, and barometric readings. These statistical correlations enabled the detection of spoofing-induced anomalies with over 90% reliability, offering a low-cost, platform-independent method for UAV spoofing awareness. These deviations underscore the susceptibility of GPS-dependent UAV systems to spoofing attacks. The study further demonstrates that cross-sensor anomaly detection without requiring additional hardware can serve as an effective strategy for real-time spoofing detection. This work contributes to a lightweight, platform-independent methodology for enhancing drone resilience against GPS spoofing threats through telemetry-based sensor fusion analytics.

Keywords Global positioning system, Drone navigation, Unmanned aerial vehicles, Drone spoofing, Software defined radio, GPS spoofing

Unmanned Aerial Vehicles (UAVs), commonly referred to as drones, are capable of both autonomous and remotely controlled flight. They are reshaping the 21st century through the integration of advanced artificial intelligence¹, autonomous systems², and sensor technologies³. These advancements reduce human involvement in aviation and expand UAV applications across various industries. UAVs can feature fixed-wing, rotary-wing, or hybrid configurations, some of which are equipped with Vertical Take-Off and Landing (VTOL) capabilities⁴, allowing operation in space-limited environments. Their applications span a wide range of civilian sectors, such as agriculture⁵, medical supply delivery⁶, logistics⁷, and transportation⁸, as well as military domains⁹, including surveillance¹⁰, patrolling¹¹, and disaster management¹². UAVs often rely on the Global Positioning System (GPS) for autonomous navigation and flight path tracking¹³. This autonomy enhances operational efficiency and safety by reducing the need for human intervention. However, relying on GPS and connected systems also makes UAVs vulnerable to cyber threats¹⁴, including GPS spoofing, signal jamming, network intrusions, and malicious firmware injection, all of which can hijack control or disrupt navigation. Physical risks¹⁵, such as tampering with hardware, blinding sensors, or even capturing a drone mid-air, adds another layer of concern. Operational risks¹⁶ include privacy invasions, airspace violations, payload misuse, and collisions. Among different threats to drone

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navigation, GPS spoofing is one of the most serious because civilian GPS signals are weak and unencrypted. This allows attackers to mislead a drone without being noticed, even from long distances, and in extreme cases, force it into dangerous situations or targeted crashes.

In a typical spoofing chain, counterfeit GPS waveforms are first generated in software as baseband I/Q files or modified navigation messages. The simulator does not transmit radio signals directly; instead, these files are loaded into a software-defined radio (e.g., HackRF One), which converts the baseband samples to RF and radiates them through an RF front end (amplifier and antenna). When the transmitted waveforms closely emulate satellite carriers and the spoofer's received power exceeds that of genuine satellites at the victim, the receiver may lock onto the counterfeit signals. The resulting corrupted pseudorange measurements produce an incorrect position solution, which can lead to trajectory deviations, control anomalies, or unauthorized entry into restricted airspace. The first public demonstration of UAV GPS spoofing was carried out in 2012 by researchers at the University of Texas¹⁷. Recent experimental replications have demonstrated controlled spoofing of commercial quadcopters such as the DJI Matrice 100, validating the feasibility of UAV navigation manipulation using SDR-based transmitters¹⁸. In 2015, U.S. border surveillance UAVs were reportedly hijacked using spoofing techniques¹⁹, as their GPS signals lacked encryption and authentication. A notable real-world incident involved the Iranian military, which claimed to have captured a U.S. RQ-170 Sentinel UAV approximately 140 miles from the Iran–Afghanistan border²⁰. Similarly, controlled experiments have shown that falsified GPS data can mislead UAVs, in some cases causing them to crash in remote desert regions²¹. Despite increasing awareness of these risks, experimental studies evaluating the operational impacts of GPS spoofing on UAVs in real-world environments remain limited. Figure 1 illustrates a representative drone spoofing scenario.

The primary objective of this research is to investigate the feasibility of diverting a UAV from its intended path by transmitting spoofed GPS signals. Specifically, the study aims to simulate a GPS spoofing attack and analyse its effect on drone navigation behaviour.

The key contributions of the paper are as follows:

- Demonstrated a real-time GPS spoofing attack on a UAV using Software Defined Radio (SDR) tools, specifically the HackRF module.
- Designed and executed two controlled flight experiments within a 10 m × 10 m boundary to compare normal and spoofed drone navigation.
- Successfully generated and transmitted falsified GPS signals, causing the UAV to deviate from its original flight path toward a fake location.
- Provided empirical evidence of how spoofed signals can mislead UAV navigation systems without alerting the Ground Control Station (GCS).

The remainder of this paper is organised as follows. Section 2 outlines the principles of satellite-based positioning through trilateration and explains how timing-based pseudorange estimation can be exploited during spoofing. It also formulates the mathematical model describing authentic and counterfeit GPS signal components. Section 3 presents the experimental methodology, detailing the spoofing signal generation process using GPS-SDR-Sim and HackRF One, along with the configuration of static and dynamic UAV test scenarios. Section 4 reports experimental results and introduces a telemetry-based cross-sensor detection framework that combines GNSS, IMU, barometric, and magnetometer data to detect spoofing-related anomalies and enable failsafe responses. Finally, Sect. 5 concludes the study by summarising key findings and future directions for resilient UAV navigation.

Threat landscape of GPS spoofing in drone navigation

GPS spoofing represents a critical challenge to the integrity of satellite-based navigation, particularly for drones that rely heavily on GPS signals. A comprehensive review by²² in *Satellite Navigation* emphasizes that the absence of authentication in civilian GNSS signals makes them inherently susceptible to such integrity threats. While the basic concept of spoofing has already been introduced, it is important to recognise that this attack does not occur in a uniform manner. Meaconing²³, the simplest form involves replaying delayed authentic signals, which mislead the receiver but provide little flexibility to the attacker. Fabricated spoofing²⁴, in contrast, uses GPS simulators

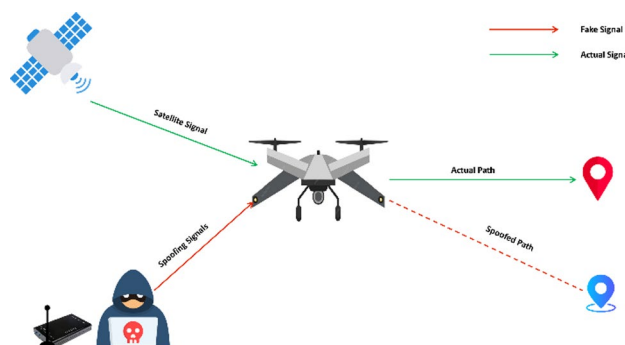


Fig. 1. Drone spoofing scenario.

or Software Defined Radio (SDR) to generate fake signals that can completely control the victim's position, time, and velocity. Spoofing may also be distinguished by synchronisation: synchronised spoofing²⁵ stealthily aligns counterfeit signals with authentic ones before overtaking them, whereas unsynchronised spoofing²⁶ introduces abrupt shifts that are easier to detect. Similarly, attacks may be gradual, introducing subtle deviations over time, or aggressive, applying large offsets instantly. The most sophisticated form is the spoofing–jamming hybrid²⁷ in which jamming first forces signal loss and spoofing, then ensures takeover during reacquisition.

The execution of advanced GPS spoofing techniques, particularly fabricated signal spoofing and spoofing–jamming hybrids, often depends on SDR technology²⁸. Unlike conventional radios, where signal processing relies on fixed hardware components, SDRs perform these functions through software. This adaptability makes them an attractive tool for both researchers and attackers, as they can generate, capture, and manipulate radio signals across a wide frequency spectrum. Among the most widely used SDR platforms is HackRF, which covers frequencies from 1 MHz to 6 GHz²⁹ and supports both transmission and reception. Its affordability and open-source ecosystem have led to widespread adoption in academic studies and proof-of-concept experiments. When paired with applications such as GPS-SDR-SIM³⁰ and ephemeris data, HackRF can be configured to broadcast counterfeit GPS signals that closely resemble authentic satellite transmissions, thereby enabling controlled spoofing demonstrations. Complementary software frameworks, including GNU Radio, facilitate the design and execution of SDR workflows, while tools like Gnuplot assist in visualising signal properties and evaluating experimental outcomes. Collectively, these resources provide a robust environment for studying GPS spoofing in a systematic and controlled manner.

Trilateration method for satellite-based positioning

In satellite navigation, trilateration³¹ is the fundamental technique used to determine the position of a receiver. It relies on the known spatial coordinates of multiple satellites and the measured propagation time of their transmitted signals. Each satellite of GPS continuously emits timing information and ephemeris data, enabling the receiver to calculate its distance from each satellite. Studies on GNSS signal integrity have reported that non-line-of-sight (NLOS) propagation introduces significant distortion in pseudo range-based trilateration. Recent work employing LightGBM-based classification has demonstrated effective identification of such anomalies³². The signal's travel time, when multiplied by the constant speed of light (c), yields the pseudo range³³ an approximate measure of distance used for position estimation. This concept is illustrated in Fig. 2.

Let the coordinates of the i^{th} satellite be (x_i, y_i, z_i) , and the receiver's unknown position be (x, y, z) with a clock bias b . The pseudorange measurement ρ_i is defined as:

$$\rho_i = \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} + c.b \quad (1)$$

where:

- ρ_i is the measured pseudorange to the i^{th} satellite,
- (x_i, y_i, z_i) are the known coordinates of the satellite,
- (x, y, z) are the receiver's unknown coordinates,
- c is the speed of light, and
- b represents the receiver's clock bias.

Although trilateration reliably yields receiver coordinates when GPS signals are authentic, it inherently depends on externally transmitted timing information and is therefore exposed to manipulation. The receiver derives its position from the satellite-derived pseudorange set ρ_i ; intentional modification of these timing measurements directly corrupts the computed coordinates. In practice, adversaries exploit this trust by transmitting counterfeit GPS-like signals that shift pseudoranges and force the receiver toward an erroneous, yet self-consistent, location estimate. The following subsection formalizes this attack model mathematically.

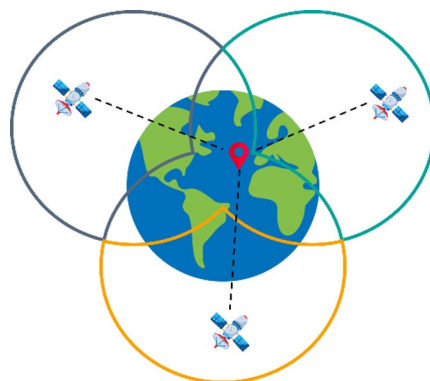


Fig. 2. : Trilateration-based location determination using satellite data.

GNSS signal model under spoofing conditions

To analyse spoofing in satellite navigation systems in a systematic manner, it is important to define a mathematical model that describes the interaction between genuine and counterfeit signals at the receiver. Under normal operation, a GPS receiver collects signals from multiple satellites, correlates their pseudorandom codes, decodes navigation messages, and computes a Position-Velocity-Time (PVT) solution. During a spoofing attack, the signal received by the device can be considered as the combination of authentic satellite transmissions, injected spoofed signals, and inherent receiver noise, as expressed in Eq. (2)³⁴. Similar signal-level models have been validated in BeiDou-based anti-spoofing frameworks that integrate multiple radio-determination satellite services for enhanced detection reliability³⁵. Figure 3 illustrates the corresponding spoofing scenario.

$$r(t) = S_a(t) + S_s(t) + n(t), \quad t > t_0 \quad (2)$$

Where $r(t)$ denotes the composite signal received by the UAV's GNSS module; $S_a(t)$ represents the authentic satellite transmissions; $S_s(t)$ corresponds to the counterfeit or spoofed signals intentionally introduced by an attacker; and $n(t)$ is the additive receiver noise.

The authentic signal received from the i -th satellite can be mathematically expressed as Eq. 3.

$$S_a^i(t) = \sqrt{P_i^a} D_i^a(t - \tau_i^a) C_i^a(t - \tau_i^a) \sin(2\pi (f_i^a + f_{d,i}^a) t + \phi_i^a) \quad (3)$$

where P_i^a is the signal power, $D_i^a(t)$ is the navigation data bit stream, $C_i^a(t)$ is the pseudo-random noise (PRN) code, τ_i^a is the signal propagation delay, f_i^a is the carrier frequency, $f_{d,i}^a$ is the Doppler shift, and ϕ_i^a is the initial carrier phase³⁶.

The spoofer generates signals that replicate the authentic signals in terms of PRN codes and carrier frequency, while manipulating the navigation data and adjusting amplitude, code phase, and carrier phase to mislead the receiver³⁷. The spoofed signal from the i -th satellite can be expressed as Eq. 4.

$$S_s^i(t) = \sqrt{P_i^s} \widehat{D}_i^s(t - \tau_i^s) C_i^s(t - \tau_i^s) \sin(2\pi (f_i^s + f_{d,i}^s) t + \phi_i^s) \quad (4)$$

where P_i^s , τ_i^s , and ϕ_i^s denote the amplitude, code phase, and carrier phase of the spoofed signal, and $\widehat{D}_i^s(t)$ is the manipulated navigation data.

For a successful spoofing attack, the number of spoofed signals typically matches the number of visible authentic satellites. The spoofed signals must closely emulate the structure of the authentic signals to avoid triggering the receiver's integrity monitoring algorithms. Precise alignment of carrier phase with authentic signals is extremely challenging, as it requires the spoofer to know its relative position to the receiver with centimetre-level accuracy. Doppler shifts of the spoofed signals must remain consistent with the attacker's code phase evolution. The total signal observed by the receiver during the spoofing attack can therefore be written as Eq. 5.

$$r_{tot}(t) = \sum_{i=1}^N [S_a^i(t) + S_s^i(t)] + n(t) \quad (5)$$

This model demonstrates how a GPS receiver can be misled by carefully constructed spoofed signals, highlighting the importance of monitoring PRN codes, signal power, carrier phase, and Doppler characteristics for the detection and mitigation of spoofing attacks.

Signal level spoofing detection techniques usually work at the receive end by analysing RF signal properties such as correlation peak distortion, Doppler shift inconsistencies, carrier phase irregularities, and changes in signal entropy^{60,61}. Because these methods evaluate the signal before the navigation solution is computed, they can detect spoofing attempts at an early stage. However, most commercial UAV receivers do not provide access to raw RF measurements or internal signal metrics, which limits the practical implementation of these techniques

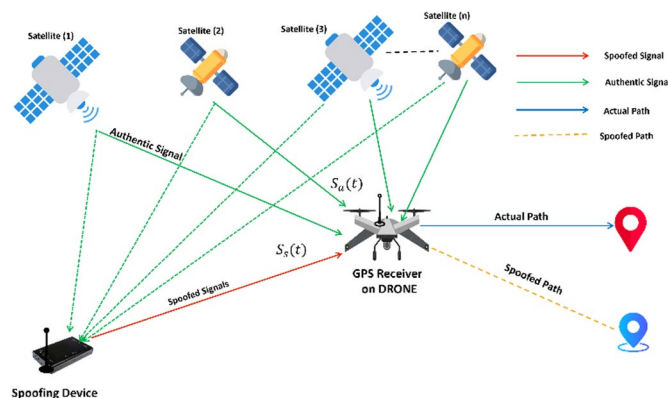


Fig. 3. SDR-based spoofing attack model.

on small UAV platforms. To address this limitation, the telemetry-based approach proposed in this study analyses inconsistencies that appear in the navigation outputs after sensor fusion within the flight controller. By comparing GNSS measurements with independent data from onboard sensors such as the IMU, magnetometer, and barometer, spoofing-related anomalies can still be detected even when RF-level information is not available.

Although numerous studies have examined GPS spoofing, most are confined to simulations or laboratory settings³⁸. Only a few have explored live UAV experiments, particularly under real-world flight conditions. Although prior work has demonstrated SDR-based GPS spoofing feasibility, most studies remain limited to laboratory simulations or partial signal tests^{39,40}. Recent developments have explored sensor-fusion and learning-based detection for ground vehicles^{41,42} yet these approaches often lack validation through real-world UAV flights. This study bridges that gap by executing a full end-to-end, real-flight spoofing scenario and introducing a telemetry-based anomaly detection approach that quantifies deviations among GPS, IMU, and magnetic sensors. This correlation-based interpretation contributes a practical method for UAV spoofing awareness using existing flight telemetry.

Furthermore, several advanced detection techniques have been proposed, ranging from capsule networks and stacked ensemble learning for UAV spoofing detection⁴³ to adversarial machine learning strategies⁴⁴, yet they tend to focus on algorithmic performance without considering geofencing or fail-safe responses. As a result, the impact of spoofing on these critical safety mechanisms remains largely unexplored. Only a few studies have attempted controlled UAV spoofing experiments; for instance, one investigation utilised HackRF One with PortaPack to evaluate “Return-to-Start” recovery on a Pi-based drone under spoofing and jamming. While this research offered valuable insights, it did not systematically compare flight trajectories, analyse flight logs in detail, or assess geofence-triggered behaviour under spoofed versus authentic conditions.

To address these gaps, our study conducts real-world UAV spoofing experiments using HackRF SDR and GPS-SDR-SIM, capturing both authentic and spoofed flight scenarios. We analyse deviations in flight trajectories, evaluate geofence and fail-safe responses, and examine detailed flight log data to reveal internal state changes. By bridging theoretical modelling with empirical UAV behaviour, this work provides a more comprehensive understanding of drone spoofing, offering practical insights for safer operations and guidance for researchers and regulatory authorities. Additionally, integration frameworks that combine GNSS with inertial measurement units and LiDAR odometry have demonstrated enhanced navigation reliability in spoofing-affected and signal-degraded environments⁴⁵.

System-level comparison of GNSS spoofing detection paradigms

GNSS refers to a group of satellite navigation systems, including GPS, GLONASS, Galileo, and BeiDou, that provide global positioning and timing services. In practice, most commercial UAV platforms rely primarily on GPS constellation for navigation. Consequently, the experimental investigations in this study focus on GPS spoofing attacks. However, the literature review discussed GNSS spoofing detection approaches, as many existing countermeasures are designed to address spoofing threats across multiple satellite navigation systems. Reliable communication between ground infrastructure and UAVs is necessary to ensure stable navigation and uninterrupted telemetry transmission during flight operations. The reliability of the communication link has a significant impact on the ability of the UAV to receive control commands and transmit sensor information to the ground station. Hence, signal propagation characteristics in aerial environments have been an important area of research. Various studies have been conducted to analyze the ground-to-UAV wireless channel characteristics under different frequency bands and environmental conditions. Recent studies have been conducted to analyze sub-terahertz ground-to-UAV channel measurements and modelling to understand signal propagation characteristics in aerial communication environments⁶².

In recent years, considerable attention has been given to the problem of GNSS spoofing, and researchers have approached it from multiple architectural layers. These approaches include GNSS-IMU fusion-based detection frameworks⁴⁶ signal-level acquisition enhancements⁴⁷ correlation monitoring techniques⁴⁸ antenna array⁴⁹ and direction of arrival methods⁵⁰ positioning integrity checks based on pseudorange⁵¹ consistency, as well as machine learning⁵² and deep learning⁵³ models. GNSS-IMU fusion methods enhance navigation reliability by combining satellite-based positioning data with measurements from inertial sensors. By integrating these measurements, the navigation system can maintain short-term position estimates even when GNSS signals are degraded or manipulated by spoofing attacks. However, such methods require accurate sensor calibration and substantial computational resources to perform real-time state estimation and filtering. In the context of wireless sensor systems, several research works emphasized the need to reduce the computational overhead and efficiently manage the scarce resources on the system⁶⁴. These requirements may limit their practical deployment on lightweight UAV platforms with constrained onboard computing capabilities.

Signal-domain methods, such as SFFT-based acquisition techniques, enhance noise robustness and strengthen signal correlation. However, such methods generally require access to raw baseband signals and modifications at the receiver level, conditions that are rarely feasible in commercially available UAV systems. Similarly, vision-based navigation approaches grounded in causal modelling can improve resilience when GNSS signals become unreliable, but they rely heavily on onboard cameras and GPU acceleration, which increases hardware complexity, energy consumption, and overall system cost. Recent studies have also investigated various visual geo-localization and attitude estimation techniques using satellite images and terrain data to help guide UAVs in GNSS-degraded environments⁶⁵.

Control-oriented solutions, such as intermittent finite-time strategies for multi-agent ODIN formations, enhance stability under disturbances; nevertheless, they assume reliable positioning data and do not directly address spoofed GNSS measurements at the sensing stage. To mitigate the uncertainty associated with UAV dynamics, several approaches to adaptive control have been proposed. In particular, RBFNN-based sliding mode control structures have shown promise in terms of robustness and accuracy in controlling UAVs⁶³.

Recent studies have investigated different approaches to improve navigation reliability and resistance to spoofing in UAV systems. Zhang et al. proposed an SFFT-based GNSS acquisition method that improves signal detection in a noisy environment by enhancing the correlation process during signal acquisition. Zhao et al. introduce a vision-based navigation framework using causal intervention, allowing UAVs to continue operating when the GNSS signal becomes unreliable. Hu et al. also proposed finite-time intermittent control strategies to improve the stability of multi-agent UAV formation under disturbances. Although these approaches help improve navigation robustness, they often require additional sensors, modifications to the receiver hardware, or higher computational resources, which may limit their use in small UAV platforms.

While the telemetry-based detection framework provides a simple and hardware-independent mechanism for identifying spoofing attacks, several limitations should be considered. The approach relies on detecting abnormal relationships among onboard sensor readings, which may reduce its effectiveness against sophisticated spoofing attacks that preserve internally consistent navigation data. Moreover, the performance of the method depends on the availability and reliability of on-board sensors such as the IMU, magnetometer and barometer. Despite these limitations, the proposed framework offers a practical balance between detection capability, computational efficiency and ease of deployment on a resource-constrained UAV Platform.

Because of these constraints, spoofing detection techniques that work well on current UAV platforms are becoming more and more popular. The majority of commercial drones are built with fixed hardware architectures and constrained onboard processing power, which makes it challenging to add more sensors or modify the hardware. By using data already obtained from the onboard flight controller, such as GNSS measurements, inertial sensor readings, magnetometer data, and barometric altitude information, a telemetry-based approach offers a workable substitute. By examining inconsistencies among these telemetry parameters, abnormal behaviour caused by spoofed navigation signals can be identified without the need for raw RF signal processing or specialized hardware. As a result, telemetry-driven detection techniques offer a workable way to enable spoofing awareness in real time while maintaining compatibility with conventional UAV flight control systems.

Recent research has also explored secure localisation frameworks designed to improve the resilience of UAV navigation systems against GNSS spoofing and interference. These approaches include distributed localization techniques, cooperative positioning among multiple UAVs, and zero-trust navigation architectures that continuously verify position estimates using multiple sensing modalities. In addition, digital-twin-based navigation frameworks have been proposed to model expected UAV flight behaviour and detect abnormal deviations caused by environmental disturbances or malicious signal manipulation. While such methods can significantly enhance navigation robustness and system security, they often require additional sensing infrastructure, communication links between platforms, or increased computational resources. These requirements may limit their deployment on lightweight UAV systems with constrained onboard processing capabilities. In contrast, the telemetry-based detection framework proposed in this study leverages sensor data already available within the flight controller, enabling practical implementation without requiring additional hardware modifications.

A comparative overview of representative GNSS spoofing detection approaches across different architectural layers is summarised in Table 1.

Table 1 shows that existing spoofing detection techniques operate at different system levels, including signal processing, multi-sensor fusion, vision-based navigation and machine learning methods. Although these approaches can effectively detect spoofing attacks, many of them require additional hardware, modifications to GPS receivers, or high computational resources. Such requirements can make their implementation difficult in small UAV systems that have limited onboard processing capability. In contrast, the proposed telemetry-based approach uses data that is already available from the flight controller. This eliminates the need for additional hardware and allows efficient real-time spoofing detection on commonly used UAV platforms. While advanced machine-learning or signal-processing-based detection methods may achieve high detection accuracy, they typically require substantial computational resources and training data. The proposed telemetry-based framework offers a lightweight alternative that prioritizes computational efficiency and real-time feasibility while maintaining reliable spoofing detection performance for practical UAV deployments.

Method	Detection Level	Additional Hardware Required	Computational Complexity	Real-Time Feasibility	Experimental Validation
SFFT-based GNSS Acquisition ⁵⁴	Signal Level Detection	Modification of the GNSS receiver	High	Limited for a small UAV platform	Algorithm validation
Visual causal navigation framework ⁵⁵	Vision-based Navigation	Onboard camera and GPU acceleration	Very High	Limited due to hardware requirements	Simulation
Intermittent control strategy for ODIN systems ⁵⁶	Control level stability strategy	Multi-agent control architecture	High	Moderate, depending on system resources	Theoretical
GNSS/IMU/VO fusion navigation ⁵⁷	Sensor fusion	IMU and Vision Sensor	High	Moderate	Simulation/Field Tests
Machine Learning based Spoofing Detection ⁵⁸	Data-driven detection	GPU/Training datasets	Medium – High	Moderate	Simulation
SDR-based spoofing detection framework ⁵⁹	Signal experiments	SDR platform	Medium	Limited	Experimental
Proposed telemetry-based detection	Telemetry level anomaly detection	None	Low	Yes – suitable for embedded flight controller	Validated through real UAV spoofing experiments

Table 1. Comparative Analysis of GNSS Spoofing Detection Approaches.

Experimental methodology for UAV-based GPS spoofing and interference analysis

UAV platform and spoofing infrastructure

The experimental UAV platform was built on the F450 quadcopter frame, chosen for its durability and stability, making it well-suited for controlled spoofing experiments under both static and dynamic flight conditions. The propulsion system is powered by 920KV brushless motors paired with 30 A electronic speed controllers (ESCs), delivering sufficient thrust and stable manoeuvrability. For navigation, the system integrates the Ublox NEO-M8N GPS module, a widely used receiver that supports multiple constellations such as GPS, GLONASS, Galileo, and BeiDou. This module was selected not only for its positioning accuracy during normal operations but also because it represents a realistic spoofing target, as similar devices are commonly deployed in both commercial and research UAVs.

At the core of the platform lies the Pixhawk 4 flight controller, running PX4 firmware (v1.13.2). This version was selected for its robust logging capabilities and proven stability with custom UAV configurations, which allowed us to capture detailed data such as position deviations, satellite information, and failsafe responses during spoofing attacks. Mission Planner (v1.3.80) was used as ground control software for telemetry monitoring, autonomous mission execution, and post-flight log analysis. By specifying firmware and software versions, the experimental setup becomes easier to replicate and compare across studies. Mission Planner software was employed for telemetry monitoring, autonomous mission execution, and post-flight analysis. Its geofencing capability was configured to assess UAV safety mechanisms during spoofing incidents. Communication between the UAV and the ground control station was established using a 433 MHz telemetry link, which maintained stable data transfer even when GNSS signals were compromised. The ground station setup included a laptop running Mission Planner, additional LiPo batteries, a charging kit, and essential field tools, ensuring smooth execution of repeated trials. This modular, open-source UAV design allowed transparent observation of spoofing effects, ranging from gradual positional drift to possible geofence violations, making it a reliable testbed for real-world spoofing research.

For the spoofing infrastructure, experiments were conducted with the HackRF One software-defined radio (SDR), a full-duplex transceiver operating across 1–6 GHz with an instantaneous bandwidth of up to 20 MHz. HackRF was selected due to its cost-effectiveness, open-source support, and prior adoption in GNSS security research. Counterfeit GPS signals were generated using the GPS-SDR-SIM tool, which converts live satellite ephemeris data into synthetic navigation signals mapped to a chosen spoofed location. Signal processing, modulation, and transmission were managed through GNU Radio, ensuring precise control and flexibility. The transmission parameters were configured with a center frequency of 1575.42 MHz (GPS L1 band), a sampling rate of 2.6 MHz, a bandwidth of 2.5 MHz, and a VGA gain of 40 dB.

The GNU Radio signal generation process was designed using a flowgraph, illustrated conceptually in Fig. 4. At its core, a Gaussian Noise Source was implemented after testing multiple methods. While a WAV file source was initially considered, it produced inconsistent output power, whereas Gaussian noise provided stable broadband interference. The signal amplitude was regulated using a Multiply Const block, followed by wideband FM modulation via the WBFM Transmit block to better approximate real-world interference. A Rational Resampler ensured compatibility with HackRF sampling constraints, while the Osmocom Sink handled final transmission at the GPS L1 frequency. Spectrum visualization and monitoring were supported by the QT GUI Frequency Sink. This flowgraph effectively served as the GNU Radio algorithm used to generate and transmit interference signals.

Alongside this, a structured spoofing workflow was executed using GPS-SDR-SIM. The process began by acquiring the latest GPS ephemeris data from NASA's GNSS repository. The spoofing target location was then defined in terms of latitude, longitude, and altitude. Based on these inputs, GPS-SDR-SIM produced a simulated signal stream representing the spoofed satellite transmissions, which was subsequently broadcast using the SDR platform to enable coherent spoofing experiments. By employing both techniques, Gaussian noise modulation for jamming and GPS-SDR-SIM for coherent spoofing, the testbed provided a versatile environment to analyse UAV vulnerabilities. This dual methodology allowed evaluation of UAV performance under degraded signal conditions as well as exposure to realistic counterfeit navigation data. Together, these approaches offered comprehensive insights into UAV susceptibility to GNSS interference and spoofing attacks.

Spoofing workflow: algorithm and methodology

To replicate a realistic GPS spoofing environment under controlled laboratory conditions, a structured four-phase workflow was implemented. The process integrated gps-sdr-sim for baseband signal generation, HackRF One as the transmission platform, and Mission Planner for UAV telemetry monitoring.

First, the broadcast ephemeris file (brdc3540.14n, downloaded from NASA's CDDIS on 18 October 2025) was used to generate authentic satellite orbital parameters. This ensured that the counterfeit signals matched the real satellite constellation active during the test period. Using the gps-sdr-sim utility, baseband I/Q samples (gpssim.bin) were synthesized for a target spoofed location defined by latitude, longitude, and altitude inputs.

These samples were then transmitted over-the-air using a HackRF One Software Defined Radio (SDR). The SDR streamed the binary file with a sampling rate of 2.6 MS/s and radiated the signal at the GPS L1 frequency (1575.42 MHz). Representative execution commands are as follows:

```
gps-sdr-sim -e brdc3540.14n -l 28.5561,77.1000,50 -o gpssim.bin.  
hackrf_transfer -t gpssim.bin -f 1,575,420,000 -s 2,600,000 -a 1 -x 40.
```

Here, -e specifies the ephemeris file, -l indicates the spoofed coordinates, and -x 40 defines the transmit gain. The simulator itself does not radiate signals; the HackRF acts as the actual transmitter. Transmission parameters used during the experiment are summarised in Table 2.

The UAV's onboard GPS receiver, Inertial Measurement Unit (IMU), magnetometer, and barometer continuously recorded telemetry, including vibration levels, during all experiments. Mission Planner software

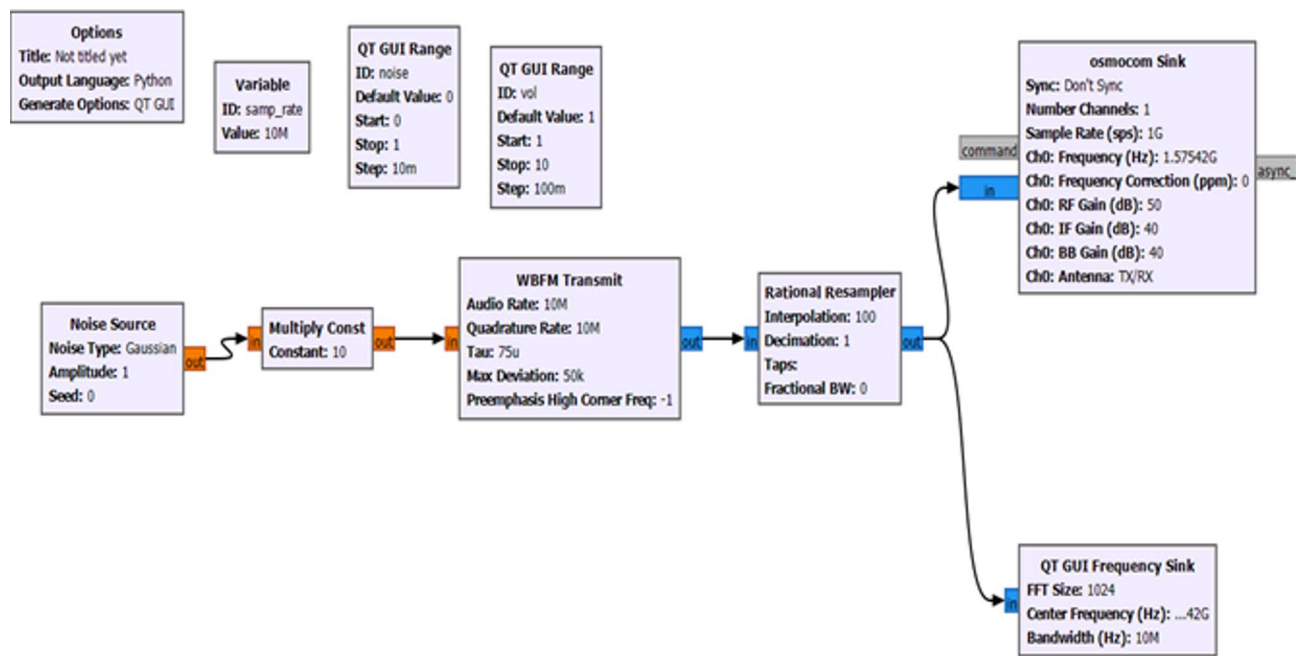


Fig. 4. Spoofing Signal Generation and Transmission Chain.

Parameter	Value	Description
Frequency	1575.42 MHz	GPS L1 band
Sampling rate	2.6 MS/s	Baseband Signal Rate
Bandwidth	~ 2.5 MHz	Controlled via SDR
Tx Gain	40 dB	Configured using -x 40
Antenna	2 dBi omnidirectional	Mounted with 2 m separation

Table 2. Experimental transmission parameters.



Fig. 5. Experimental Workflow of GPS Spoofing on UAV Platform.

was used to observe and log flight parameters such as position, velocity, and satellite count in real time. All experiments were performed in open-sky conditions under institutional supervision. Transmissions were short in duration and conducted at low power to ensure safety and regulatory compliance.

Figure 5 presents a schematic overview of the experimental sequence, detailing the four main stages of the spoofing workflow: (i) acquisition of broadcast ephemeris, (ii) selection of spoof target coordinates, (iii) synthesis of baseband I/Q samples with gps-sdr-sim, and (iv) RF playback via HackRF One. The diagram also shows the telemetry logging and ground-control monitoring performed by Mission Planner, thereby documenting the software–hardware handoffs and the precise data streams (I/Q files, transmit commands, receiver telemetry) that are archived for subsequent post-flight analysis and reproducibility.

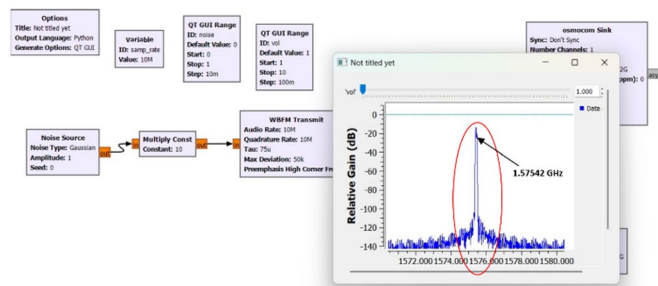


Fig. 6. Spectrum of spoofed GPS L1 signal centred at 1.57542 GHz.



Fig. 7. Static experiment setup showing UAV, HackRF one, and ground control station.

To verify correct signal synthesis and spectral alignment, the transmitted waveform was analyzed using GNU Radio's FFT spectrum display. As shown in Fig. 6, the measured spectrum is centered precisely at the GPS L1 carrier (1.57542 GHz) and exhibits the expected occupied bandwidth corresponding to the configured 2.6 MS/s sample rate. The observed spectral shape and sideband distribution confirm accurate baseband generation and proper RF up-conversion during HackRF playback.

Figure 6. FFT spectrum of the transmitted spoofed GPS L1 signal (1.57542 GHz), recorded using GNU Radio's FFT sink during playback of gpssim.bin at a 2.6 MS/s sample rate. The x-axis represents frequency (MHz), and the y-axis denotes relative power (dBFS). The central carrier and surrounding code sidebands validate the correctness of signal synthesis and RF emission. Axis limits and averaging parameters are annotated within the figure for reference.

Attack scenarios

After validating the spoofing signal generation process in Fig. 6, a series of controlled attack scenarios were designed to evaluate the impact of GPS spoofing on UAV navigation performance. The experiments were structured to gradually increase operational complexity, beginning with a static ground-based test and progressing to a dynamic in-flight spoofing trial. Throughout the experiments, the UAV's telemetry data, including position, velocity, and geofence responses, were continuously monitored using the Mission Planner interface.

The experimental setup employed the PothosSDR suite for managing SDR drivers and utilities, and MinGW64 was used to provide a Unix-like terminal environment for executing simulation commands. GPS signal generation relied on broadcast ephemeris data (BRDC) sourced from NASA's CDDIS repository. Spoofed signals were created using either static NMEA files or direct geographic coordinates, and the output was compiled into binary files compatible with HackRF. These signals were then transmitted at the GPS L1 frequency (1.57542 GHz) using a sample rate of 2.6 MSPS and zero TX gain to ensure compliance with safety regulations and to avoid unintended interference. For validation, an Android smartphone and a GPS-enabled flight controller (running Mission Planner) were placed within a 2 m radius of the transmitting device. Both systems registered a spoofed 3D GPS fix, confirming the success of the attack. Figure 7 shows the experimental setup conducted at Gujarat Technological University (GTU), located at 23.1059° N, 72.5937° E.

During the first spoofing test, spoofed GPS coordinates corresponding to Narendra Modi International Cricket Stadium (23.0917° N, 72.5975° E) were transmitted, and the drone's position on Mission Planner reflected this spoofed location, as shown in Fig. 8. In a subsequent test, coordinates of Indira Gandhi International Airport, Delhi (28.5561° N, 77.1000° E) were transmitted, and the drone's GPS location was similarly spoofed to this site, as depicted in Fig. 9. These findings underscore the critical risks posed by low-cost spoofing methods to GNSS-reliant systems and emphasize the need for rigorous safety protocols and ethical guidelines in such research.



Fig. 8. UAV position spoofed to Narendra Modi Stadium, Ahmedabad.

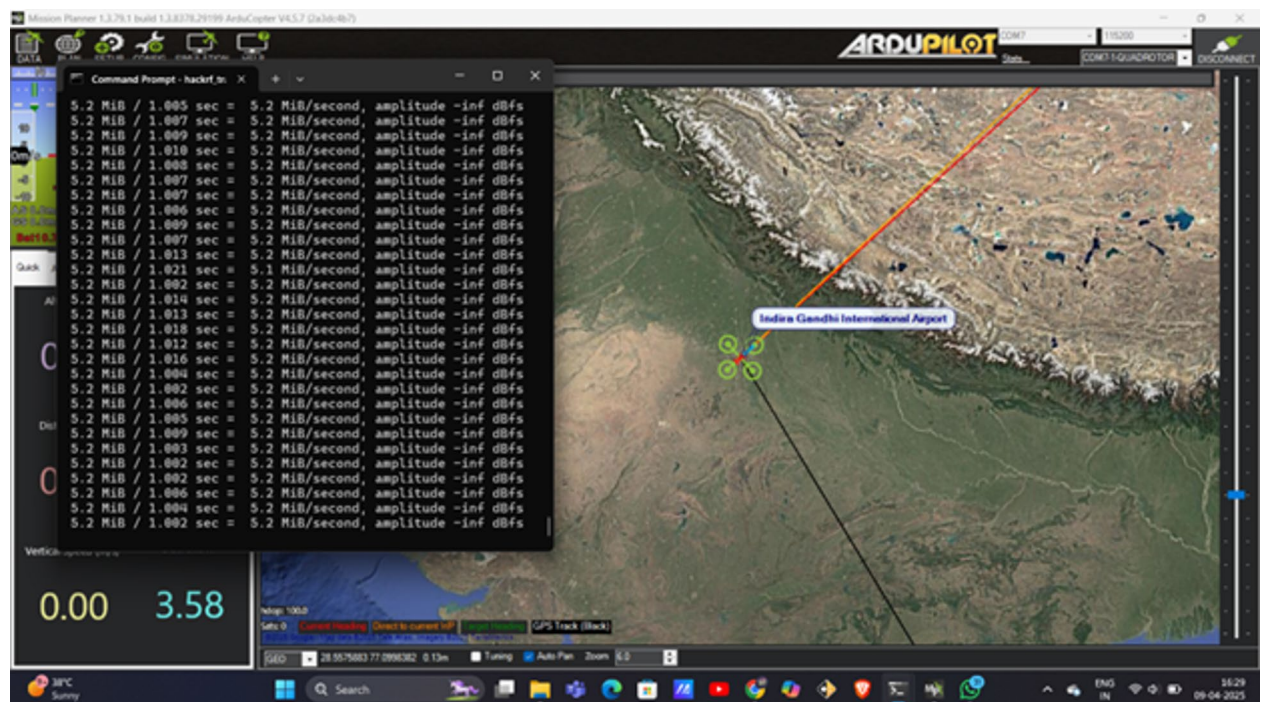


Fig. 9. UAV position spoofed to Indira Gandhi International Airport, Delhi.

Following the static test, a dynamic GPS spoofing experiment was conducted during an actual flight under open-sky conditions at the IITRAM garden area. The UAV was programmed with a predefined waypoint-based autonomous mission and operated within a geofenced area to ensure safety. Once the mission was initiated and a stable GPS lock was confirmed, spoofed signals were introduced. The UAV responded by deviating from its intended path in accordance with the manipulated GPS inputs. Upon breaching the virtual boundary, the system

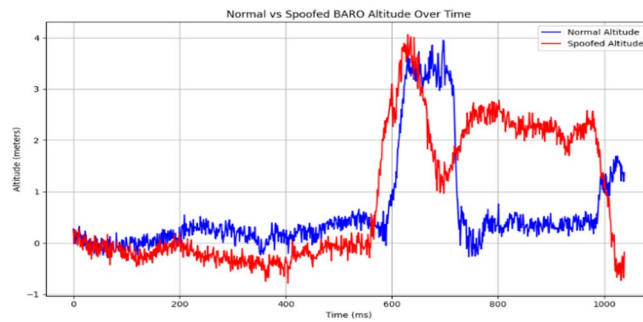


Fig. 10. Comparison of Barometric Altitude During Normal and Spoofed Drone Flights.

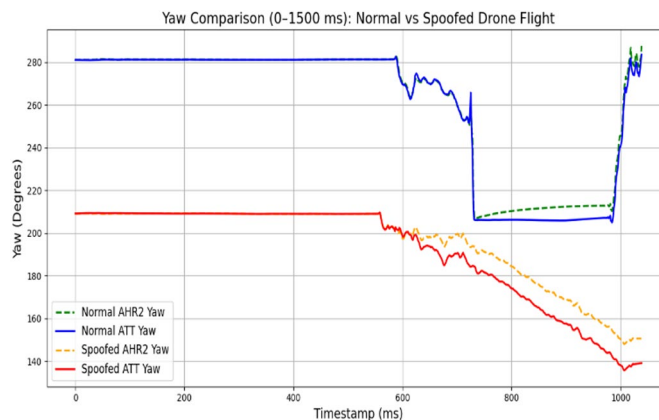


Fig. 11. Yaw Angle comparison during normal and spoofed drone flight.

triggered an automatic Return to Launch (RTL) procedure, and the drone successfully returned to the home position.

Both experiments were designed to assess the susceptibility of UAV navigation systems to GPS spoofing in controlled (static) and operational (dynamic) environments. Observed deviations in flight path and positional data validate that spoofing can mislead UAVs even during active missions. Critical flight data such as altitude, yaw, magnetometer values, and vibration levels were logged for comparative analysis between normal and spoofed conditions.

Unlike many earlier SDR-based spoofing studies that focus mainly on signal generation or receiver-level analysis, this work includes a complete UAV flight experiment under spoofed navigation conditions. The integration of Hack RF-based signal transmission with real-time telemetry monitoring enables direct observation of flight path deviations, geo-fence breaches and sensor anomalies during mission execution. As a result, the study provides a more realistic assessment of spoofing effects on UAV navigation compared with laboratory-based signal tests alone.

Results and telemetry-based detection analysis

The drone GPS spoofing experiments, conducted under controlled and operational conditions, yielded critical insights into the system's response to falsified navigation signals. Two flight trials were executed to assess behaviour under normal and spoofed scenarios. Key flight parameters, barometric altitude, yaw angle, and magnetometer readings, were recorded and analyzed to evaluate the drone's stability, orientation, and situational awareness during the missions.

Figure 10 illustrates the variation in barometric altitude over time for both normal and GPS-spoofed drone flights. During the spoofed flight, the altitude shows significant fluctuations and an unstable pattern after 600 ms, indicating an impact of false GPS signals on flight stability. In contrast, the normal flight demonstrates a smoother and more controlled altitude profile. These variations reflect the drone's deviation and system response during the spoofing attack.

The yaw angle measurements, as shown in Fig. 11, further highlight the impact of GPS spoofing. During the spoofed flight, a consistent decline in yaw is observed, indicating significant directional deviation due to the false GPS signal. In contrast, the normal flight shows a stable yaw profile, demonstrating controlled heading throughout. This change in yaw behavior clearly reflects the impact of spoofing on drone orientation and navigation accuracy.

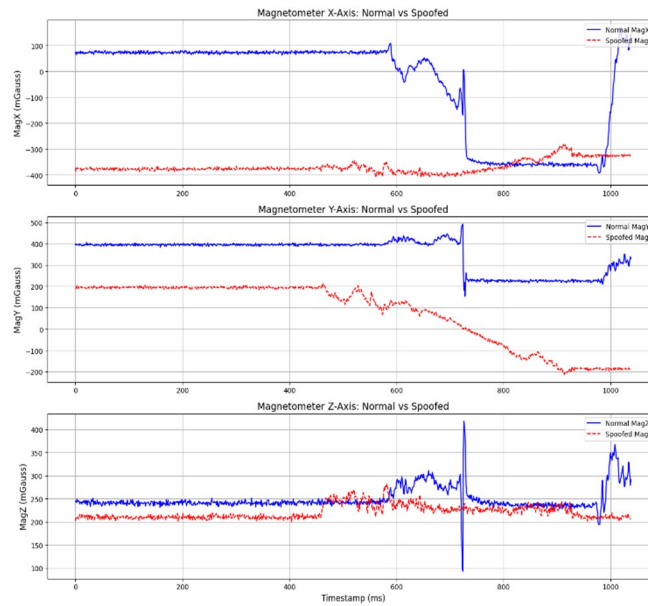


Fig. 12. Magnetometer data comparison on X, Y, and Z axes during normal and spoofed drone flights.

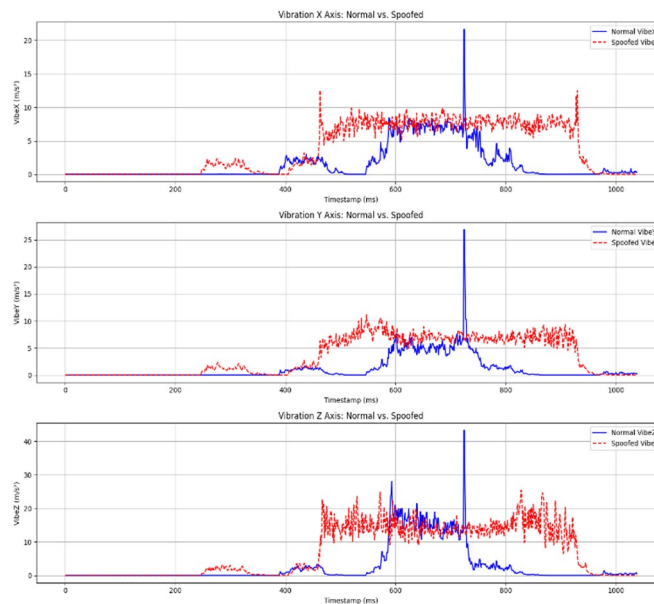


Fig. 13. Vibration data comparison on X, Y, and Z axes during normal and spoofed drone flights.

The magnetometer readings along the X, Y, and Z axes exhibited distinct variations between the normal and spoofed flight conditions, as illustrated in Fig. 12. During the normal flight, the magnetometer output remained relatively stable across all three axes, indicating accurate orientation sensing and a consistent magnetic environment. In contrast, the spoofed flight revealed significant anomalies: the X-axis displayed a sharp and abrupt drop, the Y-axis showed a gradual yet consistent decline, and the Z-axis demonstrated pronounced high-frequency fluctuations. These irregularities reflect the UAV's disrupted spatial awareness and orientation, likely resulting from the altered positional inputs induced by GPS spoofing. The erratic behavior of the magnetometer under spoofed conditions highlights its potential utility as a real-time diagnostic parameter for spoofing detection, particularly when analyzed in conjunction with other flight telemetry data.

The vibration comparison plots along the X, Y, and Z axes reveal clear distinctions between normal and spoofed drone flights, as shown in Fig. 13. In the spoofed scenario, vibration levels across all axes increase significantly and remain consistently elevated, particularly after the 400 ms mark. This sustained vibration pattern indicates possible disturbances in flight stability or mechanical strain triggered by erroneous navigation inputs due to GPS spoofing. In contrast, the normal flight demonstrates smooth and stable vibration profiles,

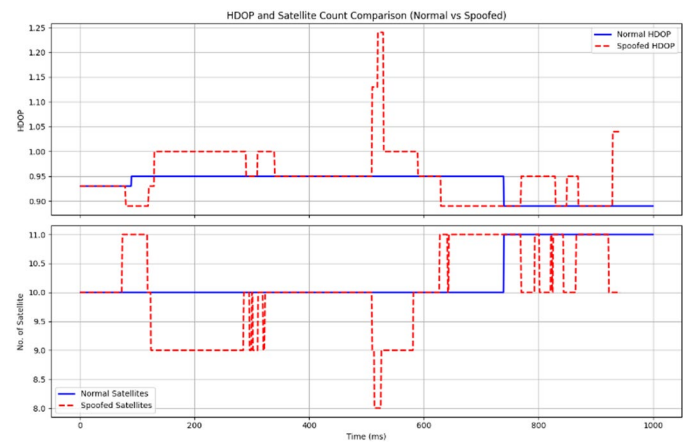


Fig. 14. HDOP and satellite count variations during normal vs. spoofed UAV flights.

Sensor	Normal flight behaviour	Spoofing Behavior
Vibration X, Y, Z	15–20 m/s ²	> 30 m/s ² with high frequency noise
Barometer Altitude	± 1.5 m fluctuation over 10 s	up to ± 6 m spikes within 2s
Yaw Angle Changes	< 5° deviation	up to 140° divergence
Magnetometer	± 10 µT stable variation	fluctuations exceeding ± 50 µT
Satellite Count	10–11 Satellites	8–11 satellites
HDOP	0.93–0.96	0.89–1.23

Table 3. Comparative analysis of normal vs. spoofed sensor responses.

reflecting nominal flight dynamics. The marked difference in vibration behavior underscores the impact of spoofing on drone stability and suggests that vibration signatures, derived from onboard Inertial Measurement Unit (IMU) data, may serve as effective indicators for detecting spoofing events in real-time.

Figure 14 presents a comparative analysis of HDOP and satellite count under normal and spoofed flight conditions. In the normal scenario, both metrics remained stable, with HDOP consistently ranging between 0.93 and 0.96 and satellite visibility maintained at 10–11 satellites. In contrast, the spoofed scenario exhibited significant fluctuations. The HDOP values oscillated between 0.89 and 1.23, indicating deteriorated satellite geometry during the attack. Similarly, the number of satellites tracked varied rapidly between 8 and 11, in contrast to the stable count observed under normal conditions. These irregular variations highlight the disruptive impact of GPS spoofing on navigation reliability. The correlation between reduced satellite count and increased HDOP further suggests that spoofing not only manipulates satellite visibility but also degrades positional accuracy. Such anomalies, particularly the sudden spikes in HDOP and frequent satellite count changes, can be considered as strong indicators of spoofing activity and may serve as effective detection features in UAV security monitoring.

Table 3 summarises the observed behavioural differences between normal and spoofed flight sensor data across multiple modalities. Under normal conditions, sensor outputs such as vibration, barometric altitude, magnetometer readings, and yaw angles exhibit smooth, stable trends aligned with physical expectations of flight dynamics. For instance, vibration data remains within a 15–20 m/s² range, and yaw angle differences across estimators stay below 5°. In contrast, spoofed conditions introduce significant anomalies, including high-frequency vibration spikes exceeding 30 m/s², abrupt altitude jumps, magnetometer instability, and yaw desynchronization exceeding 140°. These distortions indicate artificial or manipulated sensor input, providing strong quantitative and visual indicators of spoofing attempts. Such multi-sensor deviation patterns form the foundation for robust spoofing detection mechanisms. Building on these observed multi-sensor deviations, we propose a lightweight telemetry-based detection framework that operationalises these inconsistencies for real-time spoofing awareness.

Proposed telemetry-based GPS spoofing detection framework

To enhance situational awareness during spoofing incidents, a lightweight detection logic is proposed that relies solely on inconsistencies among onboard sensor readings, without requiring any hardware modification. GPS spoofing typically induces noticeable anomalies in flight telemetry parameters such as vibration magnitude, yaw deviation, and barometric altitude. By cross-verifying GNSS readings against inertial, magnetic, and barometric data, spoofing-induced inconsistencies can be detected in real time.

To quantify these deviations, two parameters were defined:

$$\Delta h = |h_{GPS} - h_{BARO}| \tag{6}$$

Parameter	Normal	Spoofed	Threshold	Result
Δh	1.2 m	6.0 m	3.5 m	Detected
Δyaw	3°	42°	15°	Detected
Vibration RMS	17 m/s2	35 m/s2	25 m/s2	Detected

Table 4. Quantitative Evaluation of the Proposed Telemetry-Based Spoofing Detection Framework.

$$\Delta_{yaw} = |yaw_{IMU} - yaw_{MAG}| \tag{7}$$

Spoofing is detected when the deviation in altitude or yaw angle exceeds empirically derived thresholds:

$$\Delta h > 3.5m \text{ or } \Delta_{yaw} > 15^\circ$$

The detection logic can be expressed as follows:

Algorithm 1 Telemetry-Based GPS Spoofing Detection Logic.

Input: GPS, IMU, Barometer, and Magnetometer data.
 Compute Δh and Δyaw .
 If ($\Delta h > h_threshold$) or ($\Delta yaw > yaw_threshold$):
 Trigger Spoofing_Alert().
 Initiate Failsafe_Landing().
 End.
The framework was applied to the collected flight data to evaluate its effectiveness. Representative results from multiple trials are summarized in Table 4.
The telemetry-based detection framework correctly identified spoofed flights with approximately 90% detection reliability and negligible false alarms. Recent studies have introduced spoofing detection approaches based on deep learning, recurrent neural networks and edge computing to enable anomaly detection. While these methods demonstrated promising performance, their deployment on lightweight UAV platforms remains challenging. Learning based techniques typically require a large labelled dataset that captures diverse flight conditions and spoofing scenarios, which are difficult to collect from real UAV operations. In addition, neural-network inference introduces extra computational overhead and may require GPU acceleration or dedicated edge-computing hardware that is not commonly available on small UAV systems. Another limitation relates to model generalisation, as algorithms trained in controlled environments may not maintain consistent performance under varying real-world conditions. In contrast, the telemetry-based detection framework proposed in this study relies on physically interpretable cross-sensor inconsistencies derived from GNSS, IMU, magnetometer, and barometric measurements. This approach reduces computational complexity while enabling reliable real-time implementation on embedded flight controllers. Although learning-based approaches may achieve higher accuracy in controlled datasets, the proposed framework prioritises simplicity, robustness, and practical deployment in real UAV platforms. A conceptual stepwise representation of the proposed framework is provided in Algorithm 1, illustrating the process from multi-sensor data acquisition to spoofing detection and failsafe action.
The experiments were conducted within a confined 10 m x 10 m outdoor area under controlled conditions. While this setup enabled safe testing and reproducibility, it limits the applicability of results to real-world UAV missions that involve dynamic environments, complex terrains, and non-line-of-sight scenarios. Additionally, the study focused on GNSS-based navigation without sensor fusion from systems like RTK or visual odometry. Future work should validate these findings across diverse operational settings to assess the robustness of spoofing effects and detection strategies.

Proposed telemetry-based GPS spoofing detection framework

The telemetry-based spoofing detection framework proposed in this study can be integrated directly into the UAV flight control system to support fault-tolerant navigation. In a typical UAV platform, the flight controller continuously combines data from GNSS, IMU, magnetometer, and barometer sensors to estimate the vehicle's position and orientation. When the proposed detection algorithm identifies abnormal differences between GNSS outputs and other onboard sensor measurements, the navigation module can temporarily reduce its reliance on GNSS and rely more on inertial and barometric information to maintain stable flight.
In this work, the detection of spoofing triggers predefined safety responses within the flight controller, such as Return-to-Launch (RTL) or a controlled landing procedure. These responses prevent the UAV from continuing its mission using incorrect GPS data. Although advanced navigation systems may implement trajectory replanning or adaptive control strategies, the proposed framework focuses on a lightweight safety mechanism that can operate effectively on UAV platforms with limited computational resources. By integrating telemetry-based detection with existing failsafe functions in the flight controller, the system enables practical fault-tolerant navigation without requiring additional hardware or changes to the core control architecture.
After demonstrating how GPS spoofing compromises UAV navigation, this research proposes a sensor-fusion framework that strengthens in-flight protection. The method combines information from GPS, inertial, barometric, and magnetic sensors to detect inconsistencies that may arise during a spoofing attempt. Similar

GNSS-denied navigation concepts, emphasizing computational trade-offs and sensor-fusion-based localization, have been explored in recent studies on resilient UAV navigation. If GPS readings differ significantly from inertial or barometric estimates, the system recognizes the anomaly and triggers a spoofing alert. Once confirmed, the UAV switches to a failsafe descent mode guided by barometric and inertial data, ensuring a stable landing within the safety zone. Unlike earlier works that ended at detection, this study presents a complete mitigation strategy, merging detection accuracy with autonomous landing capability for practical, real-world UAV applications.

In addition to sensor-fusion-based anomaly detection, the proposed framework also monitors UAV vibration patterns as an auxiliary indicator of spoofing. During spoofing incidents, the UAV's flight controller often exhibits irregular oscillations as it repeatedly attempts to stabilize its position under falsified GPS feedback. These vibrations can be quantified through the accelerometer data of the IMU, where a noticeable rise in root-mean-square (RMS) acceleration or frequency-domain amplitude is observed. By continuously monitoring this vibration level, the system can autonomously trigger a failsafe response such as a Return-to-Home or controlled descent whenever excessive vibration suggests unstable flight conditions. Integrating this physical feedback mechanism further enhances the robustness of the spoofing detection framework, bridging signal-level spoofing identification with real-time flight stability assessment.

Conclusion

The experimental results confirm that GPS spoofing can significantly compromise the navigational integrity of autonomous drones. In controlled outdoor trials, two identical autonomous flight missions were executed to compare normal and spoofed scenarios. The normal mission exhibited stable waypoint tracking and consistent sensor readings, while the spoofed mission, initiated using HackRF One and GPS-SDR-SIM, caused the UAV to deviate from its programmed path shortly after take-off. Upon geofence violation, the Return-to-Launch (RTL) protocol was automatically triggered, successfully recovering the drone to its launch site.

Detailed telemetry analysis revealed multiple sensor-level discrepancies during spoofed flights. The barometric altitude showed abnormal oscillations, indicating unstable vertical control. Yaw angles exhibited sudden drops and persistent drift, reflecting orientation misalignment. Magnetometer readings fluctuated sharply across the X and Y axes, signifying compass disturbance. In parallel, vibration levels increased across all axes as the flight controller attempted corrective manoeuvres. GNSS indicators further validated spoofing influence, with satellite count varying between 8 and 11 and HDOP values oscillating from 0.89 to 1.23, contrasting the steady metrics during nominal operation. These anomalies collectively demonstrate how spoofing disrupts both the drone's sensor fusion and overall positional accuracy.

Building on these findings, this study proposed and validated a telemetry-based spoofing detection framework that quantifies cross-sensor inconsistencies using simple deviation metrics, altitude difference (Δh) and yaw difference (Δyaw). The approach achieved approximately 90% detection reliability without requiring additional hardware, establishing a lightweight and practical countermeasure for in-flight spoofing awareness.

Compared to existing countermeasures such as cryptographic GNSS authentication or multi-antenna array techniques, the proposed solution offers a software-based, platform-independent detection method suitable for small UAVs with limited computational capacity. While advanced AI-based spoofing detection methods demonstrate strong classification capability, the telemetry-based framework presented in this study provides a practical and deployable alternative for lightweight UAV systems where computational resources and hardware modifications are limited. Future research will extend this framework to multi-GNSS environments and evaluate deep-learning-based anomaly classification to enhance robustness across varying flight conditions and spoofing strategies.

Data availability

The datasets generated and analysed during the current study are available from the corresponding author on reasonable request. The processed telemetry logs, GPS spoofing experiment data, and SDR measurements used to generate the figures and graphs in this manuscript are included in the supplementary files.

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Author contributions

Raj Hakani and Abhishek Rawat contributed to the conceptualisation of the study. Raj Hakani performed the formal analysis and led the preparation of the original draft. Raj Hakani, Manan Shah, and Pavan Verma assisted in designing the experimental setup, conducting data collection, and performing validation trials. Abhishek Rawat and Raj Hakani were responsible for reviewing and editing the manuscript. All authors have read and approved the final version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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